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Investor Composition and the Liquidity Component in the U.S. Corporate Bond Market

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1 Big picture

- **Question.** Why have secondary-market frictions become much more important for corporate bond credit spreads since the mid-2000s, even though standard liquidity measures such as bid-ask spreads did not obviously worsen? Can the change in the investor base explain this trend?
- **Setting.** The paper studies the U.S. corporate bond market from 2005Q2 to 2019Q2. It combines bond-level trading data, bond characteristics, firm fundamentals, and investor holdings to connect changes in credit spreads to changes in who holds corporate bonds.
- **Core challenge.** The level of illiquidity is not the same thing as the *price sensitivity* of bonds to illiquidity. A market can look liquid in terms of average bid-ask spreads and still become more fragile if the marginal investor trades more often and therefore cares more about secondary-market frictions.
- **Main reduced-form finding.** The loading of credit spreads on bid-ask spreads rises sharply over time. Before the Global Financial Crisis, a 100 basis point difference in bid-ask spreads was associated with about 10–20 basis points of extra credit spread for investment-grade bonds; by 2019 the number is about 60 basis points. For high-yield bonds, the corresponding effect rises from roughly 100 to 250 basis points.
- **Liquidity-component finding.** Because the loading coefficient rises so much, the median liquidity component of credit spreads increases from roughly 10% of the credit spread before the financial crisis to roughly 30% by 2019.
- **Cross-sectional mechanism.** Bonds held by shorter-term investors trade more frequently and have a stronger relation between credit spreads and bid-ask spreads. Using the discontinuity at the 10-year maturity threshold created by intermediate-term bond fund mandates, the paper shows that short-term investors causally increase secondary-market trading activity.
- **Main model idea.** The model features heterogeneous investors who differ in their frequency of liquidity shocks and heterogeneous bonds that differ in maturity and default risk. Bonds are traded in an over-the-counter secondary market with search frictions and bid-ask spreads. As the risk-free rate falls, more short-term investors enter the corporate bond market to reach for yield.
- **Main theoretical mechanism.** More short-term investors increase trading needs, so each unit of transaction cost is incurred more often before maturity. At the same time, the inflow of new investors lowers seller-buyer ratios in secondary markets. Both channels change how bid-ask spreads map into credit spreads.
- **Main quantitative result.** A calibrated version of the model matches the observed increase in bond turnover, the increase in the loading of credit spreads on bid-ask spreads, and the rise in the liquidity component. The model predicts that the liquidity component grows by about 2.5 times, close to the 2.8-fold increase in the data.
- **Economic takeaway.** The rise of mutual funds and ETFs changed the *composition* of investors in corporate bonds. That change made credit spreads more sensitive to secondary-market frictions, even though average bid-ask spreads did not rise. In other words, investor composition can make the market more fragile without making standard liquidity levels look worse.

2 Data and measurement (key objects)

2.1 Corporate bond, price, and firm-level data

- **Bond characteristics.** Bond-level characteristics come from **Mergent FISD** and the **WRDS Bond Return** database.
- **Firm fundamentals.** The bond information is merged with **CRSP** for equity prices and with **Compustat** for firm accounting information.
- **Trading data.** Credit spreads and bid-ask spreads are constructed from **TRACE**, the Enhanced Trade Reporting and Compliance Engine run by FINRA.
- **TRACE filters.** The paper excludes interdealer trades, agency trades, and retail trades (face value below \$100,000). These filters are important because the model is about frictions faced by final investors in OTC trading, not about central electronic platforms or dealer-to-dealer trades.
- **Credit spread measure.** For each bond-quarter, the credit spread is the reported TRACE yield minus the corresponding Treasury yield, using the Gurkaynak–Sack–Wright Treasury curve.
- **Bid-ask spread measure.** The bid-ask spread is the difference between the volume-weighted customer buy price and customer sell price, normalized by the mid price and then averaged over the quarter.
- **Bond sample.** The analysis focuses on U.S. corporate debentures with fixed coupons that are nonconvertible, nonputtable, and nonexchangeable.
- **Sample period.** The main period is **2005Q2–2019Q2**.

2.2 Investor holdings data and the investor-composition object

- **Investor holdings source.** Holdings come from **Lipper eMaxx** (Thomson Reuters), which contains quarterly CUSIP-level corporate bond holdings for insurance companies, mutual funds, and other investors.
- **Coverage.** eMaxx covers roughly **40% to 50%** of total amount outstanding throughout the sample. The missing holders are mostly foreign accounts and banks.
- **Key idea.** Investors differ in how frequently they reshuffle bond holdings. The paper interprets more persistent net trading as evidence that the investor is more short-term and more frequently hit by liquidity shocks.
- **Investor-level net transactions.** For fund j in quarter t ,

$$net_transaction_{j,t} = \frac{|\sum_i holding_{i,j,t} - \sum_i holding_{i,j,t-1}|}{\sum_i holding_{i,j,t-1}}.$$

- **Persistent trading component.** The paper averages this over the past four quarters:

$$NT_{j,t} = \frac{1}{4} \sum_{t'=1}^4 net_transaction_{j,t-t'}.$$

- **Bond-level investor composition.** For bond i at time t ,

$$investor_comp_{i,t} = \frac{\sum_j holding_{i,j,t} \times NT_{j,t}}{\sum_j holding_{i,j,t}}.$$

A higher value means the bond is held by shorter-term investors.

- **Cross-sectional sample.** After requiring at least 20% eMaxx coverage and applying the same TRACE filters, the bond-level holdings sample contains about **15,256 unique bonds** and **3,217 unique firms**.

2.3 Aggregate trend objects

- **Main aggregate regression.** Quarter by quarter, the paper estimates

$$CS_{i,t} = \alpha_t + \beta_t BA_{i,t} + \gamma_t' X_{i,t} + \varepsilon_{i,t},$$

where $X_{i,t}$ includes bond age, maturity, coupon, offering amount, rating fixed effects, leverage, size, profitability, equity volatility, total assets, and industry fixed effects.

- **Interpretation of β_t .** The coefficient β_t measures the cross-sectional loading of credit spreads on bid-ask spreads in quarter t .
- **Liquidity component.** For each bond,

$$liquidity\ component_{i,t} = \beta_t \times \frac{BA_{i,t}}{CS_{i,t}}.$$

The paper studies the median of this object over time.

2.4 Key empirical facts

- The loading of credit spreads on bid-ask spreads more than doubles between 2005 and 2019, for both investment-grade and high-yield bonds.
- The median liquidity component rises from around **10%** of the credit spread before the GFC to around **30%** by 2019.
- The share of mutual funds and ETFs in corporate bond holdings rises substantially over the same period, especially after 2008.
- The median number of trades rises by more than **60%** over the sample period.
- In the cross section, shorter-term investors hold bonds with **shorter maturities**, **higher default intensity**, **lower bid-ask spreads**, and **more trading activity**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Reduced-form trend specification

- The first empirical question is whether the market price of illiquidity has changed over time. This is captured by the time-varying coefficient β_t in

$$CS_{i,t} = \alpha_t + \beta_t BA_{i,t} + \gamma_t' X_{i,t} + \varepsilon_{i,t}.$$

- **Interpretation.** A rise in β_t means that a given difference in bid-ask spreads is associated with a larger difference in credit spreads. This is not just “less liquidity” in levels. It is a change in the *pricing kernel* attached to secondary-market frictions.

3.2 Cross-sectional causal design around the 10-year maturity cutoff

- The paper uses a discontinuity in investor mandates at the 10-year maturity threshold. Intermediate-term corporate bond mutual funds often restrict themselves to bonds with maturity below 10 years. As a bond crosses below 10 years to maturity, it becomes eligible for a much larger set of short-term investors.
- First stage:

$$Inv_Comp_{i,t} = \alpha + \beta_1 1_{ttm>10} + \gamma' X_{i,t} + \varepsilon_{i,t}.$$

- Second stage for trading activity:

$$Y_{i,t} = \alpha + \beta_1 \widehat{Inv_Comp}_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t},$$

where $Y_{i,t}$ is either bond turnover or the number of trades.

- Second stage for the pricing of illiquidity:

$$CS_{i,t} = \alpha + \beta_1 BA_{i,t} + \beta_2 \widehat{Inv_Comp}_{i,t} + \beta_3 BA_{i,t} \times \widehat{Inv_Comp}_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t}.$$

- **Interpretation.** The instrument isolates exogenous movement in investor composition. If the interaction coefficient β_3 is positive, then an exogenous increase in short-term investors causally strengthens the loading of credit spreads on bid-ask spreads.

3.3 Model environment

- Time is continuous. Investors are risk-neutral and infinitely lived.
- Investors are heterogeneous in their liquidity-shock frequency θ_j . A higher θ_j means the investor is more short-term and is more likely to need to sell the bond in the future.
- Bonds are heterogeneous in their maturity hazard δ_i , default intensity d_i , and recovery value s_i . Each bond has face value 1 and coupon flow r_i .
- There is a **frictionless primary market** and a **frictional secondary market**. In the secondary market, impatient sellers search for buyers and trade through dealers who extract part of the trading surplus.
- There is also a frictionless risk-free outside asset paying r_f .

3.4 Secondary-market search, bid-ask spreads, and value functions

- Let $\alpha_{s,i}$ be the measure of sellers of bond i , and let $\alpha_b(\theta)$ be the measure of buyers of type θ . Matching in a submarket follows the Cobb–Douglas function

$$m(\alpha_{s,i}, \alpha_{b,j}) = \eta \alpha_{s,i}^\gamma \alpha_{b,j}^{1-\gamma},$$

so the buyer and seller meeting rates are

$$\mu_b(\lambda) = \eta\lambda^\gamma, \quad \mu_s(\lambda) = \eta\lambda^{\gamma-1},$$

where λ is the seller-buyer ratio.

- The total bid-ask spread has an endogenous and an exogenous component:

$$P_{b,i} - P_{s,i} = \xi_i + \varepsilon_i^\xi.$$

- The endogenous component is a fraction of trading surplus:

$$\xi_i = \kappa(V_{h,i}(\theta) - V_{b,i}(\theta) - V_{s,i}).$$

- The value of a patient holder is

$$\rho V_{h,i}(\theta) = r_i + \delta_i(1 - V_{h,i}(\theta)) + d_i(s_i - V_{h,i}(\theta)) + \theta(V_{s,i} - V_{h,i}(\theta)).$$

- The value of a seller is

$$\rho V_{s,i} = r_i - \Delta + \delta_i(1 - V_{s,i}) + d_i(s_i - V_{s,i}) + \mu_s(\lambda)(P_{s,i} - V_{s,i}).$$

- The value of a buyer is

$$\rho V_{b,i}(\theta) = \mu_b(\lambda)(V_{h,i}(\theta) - V_{b,i}(\theta) - P_{b,i}).$$

- In equilibrium, buyers are indifferent between buying in the frictionless primary market and entering the secondary market:

$$V_{h,i}(\theta) - 1 = V_{b,i}(\theta).$$

- **Interpretation.** The model makes the corporate bond market a directed-search environment. Investor composition is an endogenous sorting object, not just a reduced-form characteristic.

3.5 Equilibrium characterization

- Define the effective maturity/default object

$$\tilde{\delta}_i \equiv \delta_i + d_i.$$

In the model, sorting depends on $\tilde{\delta}_i$: short maturity and high default intensity both make a bond effectively “shorter” from the perspective of future trading needs.

- **Lemma 1.** When the exogenous component of the bid-ask spread is not too large, buyers and sellers trade with the first counterparty they meet. This lets the search problem collapse to a tractable system.
- **Proposition 1. Positive assortative matching.** Short-term investors are matched with short-term and high-default-probability bonds:

$$\theta'(\delta + d) > 0.$$

- **Interpretation.** Investors with a high chance of needing to sell care more about early maturity and less about eventual default, because they may sell before maturity or default. Hence they sort into short-duration, riskier submarkets.
- **Proposition 2.** The steady-state equilibrium is characterized by a pair of ODEs for the seller-buyer ratio $\lambda(\tilde{\delta})$ and the equilibrium investor type $\theta(\tilde{\delta})$, together with cutoff conditions for market participation.
- **Corollary 1.** The endogenous component of the bid-ask spread is decreasing in $\tilde{\delta}$ and in θ . In the calibration, submarkets with shorter-term investors end up having lower bid-ask spreads, even though those same investors make credit spreads more sensitive to trading frictions.
- **Proposition 3.** When the risk-free rate falls, more short-term investors enter the illiquid bond market (reach for yield). This raises the cutoff type for bond-market participation, shifts all submarkets toward shorter-term investors, and lowers seller-buyer ratios across submarkets.
- **Interpretation.** This result is the paper’s main macro mechanism: a lower risk-free rate changes who enters the corporate bond market, and that composition change alters both trading intensity and the pricing of liquidity.

3.6 Credit spreads and the liquidity channel

- **Lemma 2. Interest-rate decomposition.** In equilibrium, bond yields have three pieces:

r_i = discount-rate component+default-spread component+secondary-market liquidity component.

- The liquidity component depends on trading frequency, matching frictions, seller-buyer ratios, and bid-ask spreads.
- **Corollary 2.** The sensitivity of the interest rate to the exogenous component of the bid-ask spread increases when more short-term investors enter the market (equivalently, when the risk-free rate falls).
- **Interpretation.** With more short-term investors, a given transaction cost matters more because the bond is expected to be traded more often before maturity.

3.7 Calibration and the decomposition of the β coefficient

- The calibration is annual. The authors focus on bonds with maturities between 2 and 15 years.
- Investor types follow an exponential distribution, and the exogenous component of bid-ask spreads is assumed normal.
- The model is calibrated separately to a 2005 economy and a 2019 economy. The key difference between the two is the participation cutoff $\bar{\theta}$: more short-term investors enter by 2019.
- The calibration targets:
 1. average bond turnover in 2005 and 2019,
 2. the loading of credit spreads on bid-ask spreads in 2005 and 2019,
 3. average credit spreads in 2005 and 2019,

4. average bid-ask spreads in 2005 and 2019, and
 5. the ratio of the standard deviation of residual bid-ask spreads to the standard deviation of residual credit spreads in 2005 and 2019.
- The paper decomposes the empirical loading

$$\beta \approx \phi_1 + \phi_2,$$

where ϕ_1 is the **direct** effect of exogenous transaction costs on credit spreads, and ϕ_2 is the **indirect** effect that comes from the correlation between bid-ask spreads and seller-buyer ratios (trade delays).

- **Interpretation.** The paper’s quantitative punchline is that the indirect, endogenous-liquidity channel is crucial. It is not enough to think of bid-ask spreads as exogenous trading fees.

4 Identification and instruments (what makes the estimates credible)

4.1 What is hard to identify

- Credit spreads co-move with default risk, dealer balance-sheet conditions, investor risk appetite, and many other macro-financial variables. So a raw relation between credit spreads and bid-ask spreads is not automatically informative about liquidity.
- **Investor composition is endogenous.** Short-term investors may prefer certain bonds exactly because those bonds are easier to trade or have different default risk.
- Bid-ask spreads themselves are endogenous in the model and are correlated with seller-buyer ratios and waiting times.

4.2 What identifies the aggregate trend

- The quarter-by-quarter cross-sectional regressions control for a rich set of bond and firm characteristics.
- The authors show robustness to alternative liquidity measures, to BBB-only subsamples, to samples with CDS spreads, and to specifications with bond fixed effects.
- These exercises do not fully establish causality, but they make it much harder to explain the rising β_t coefficient purely by changes in default risk.

4.3 What identifies the causal role of investor composition

- The key instrument is the discontinuity created when a bond crosses the 10-year maturity threshold and becomes eligible for intermediate-term mutual funds.
- The design isolates local variation in holdings by short-term investors that is driven by investment mandates rather than by contemporaneous firm fundamentals.

- Because the paper also instruments the interaction term $Inv_Comp \times BA$ with $1_{ttm>10} \times BA$, it can identify whether investor composition changes the sensitivity of credit spreads to bid-ask spreads, not just levels.

4.4 What identifies the model parameters

- The rise in aggregate bond turnover identifies the shift in the market participation cutoff toward shorter-term investors.
- The change in the coefficient on bid-ask spreads pins down how sensitive credit spreads become when the investor base shortens.
- The joint behavior of average credit spreads, average bid-ask spreads, and the residual standard deviation ratio disciplines the matching function, the liquidity-shock discount Δ , dealer surplus extraction κ , and the exogenous noise in bid-ask spreads.
- The cross-sectional sorting patterns—shorter-term investors holding shorter-maturity and riskier bonds, with more trades and lower bid-ask spreads—discipline the positive assortative matching mechanism.

4.5 Why the application is credible, and limitations

- The paper brings together unusually rich data: bond-level prices, firm fundamentals, and security-level holdings.
- The identification strategy is credible because the 10-year threshold is tied to institutional rules of a large class of mutual funds.
- Important limitations remain. The model treats liquidity shocks as exogenous, which may be inaccurate for active mutual funds that trade in response to price movements. The dealer sector is not modeled explicitly. Dealer regulation may matter independently of investor composition.
- The authors explicitly recognize that they cannot completely rule out liquidity-supply explanations such as changes in dealer constraints; instead, they show that investor composition can quantitatively account for a large share of the observed trend.

4.6 Relation to the literature

- The paper speaks to the literature on liquidity components of corporate bond spreads by showing that the *pricing* of liquidity changed over time, not just the level of liquidity.
- It contributes to the corporate bond fund fragility literature by emphasizing the role of mutual funds and ETFs as short-term investors with high turnover.
- It contributes to OTC/search-theoretic asset pricing by embedding investor heterogeneity and bond heterogeneity into a directed-search framework.
- It also connects to the clientele literature in bond markets, showing how changes in the composition of holders affect prices and yields.

5 Tables and figures

5.1 Figure 1: the loading of credit spreads on bid-ask spreads over time

Read:

- This is the core reduced-form time-series fact of the paper. The coefficient on bid-ask spreads rises steadily after the financial crisis for both investment-grade and high-yield bonds.
- For investment-grade bonds, the effect of a 100 basis point difference in bid-ask spreads rises from about 10–20 basis points before the GFC to around 60 basis points by 2019.
- For high-yield bonds, the same comparison rises from roughly 100 basis points to roughly 250 basis points.
- The figure says that even if average liquidity does not collapse, liquidity *matters more* for prices by the end of the sample.



Figure 1. Relationship between credit spreads and bid-ask spreads. We regress credit spreads on bid-ask spreads quarter by quarter for all bonds as in equation (1), controlling for bond and firm characteristics. Panel A (B) plots the coefficient in front of bid-ask spreads for investment-grade (high-yield) bonds. Standard errors are clustered at the firm level. The shaded regions indicate the 95% confidence intervals. Bid-ask spreads are winsorized at the top and bottom 1%. (Color figure can be viewed at wileyonlinelibrary.com)

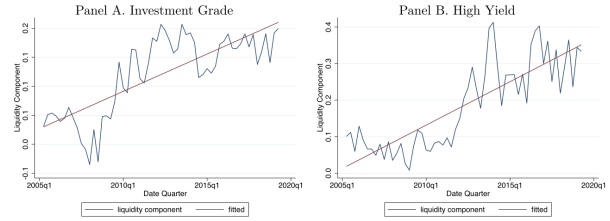


Figure 2. Liquidity component as a fraction of credit spreads. The figure plots the liquidity components of investment-grade bonds and high-yield bonds over time. For each quarter t , we first run equation (1) to get the coefficient of bid-ask spread β . Bond characteristic data are from Mergent FISD and WRDS bond returns; price data are from TRACE. Next for each bond, $liquidity_component_{i,t}$ is defined as $\frac{\beta \times bid_ask_{i,t}}{credit_spread_{i,t}}$. Panel A plots the median liquidity component of investment-grade bonds over time, and Panel B plots the corresponding trend for high-yield bonds. The red line is a fitted linear trend over time. Bid-ask spreads are winsorized at the top and bottom 1%. (Color figure can be viewed at wileyonlinelibrary.com)

5.2 Figure 2: liquidity component as a fraction of credit spreads

Read:

- Figure 2 translates the rising β_t coefficient into economic magnitudes.
- The median liquidity component rises from around 10% of the credit spread before the GFC to around 30% by 2019, for both investment-grade and high-yield bonds.
- The important point is that this rise occurs even though bid-ask spreads themselves do not rise much. The change comes mainly from the coefficient, not from the level of measured illiquidity.

5.3 Figure 3 and Table I: who holds which bonds, and what happens at the 10-year cutoff?

Read:

- Figure 3 shows the basic sorting facts: bonds held by short-term investors have shorter maturities, worse ratings, lower bid-ask spreads, and many more trades.
- Table I then gives the quasi-experimental version of the same story. Crossing below 10 years to maturity raises the short-term-investor composition measure.

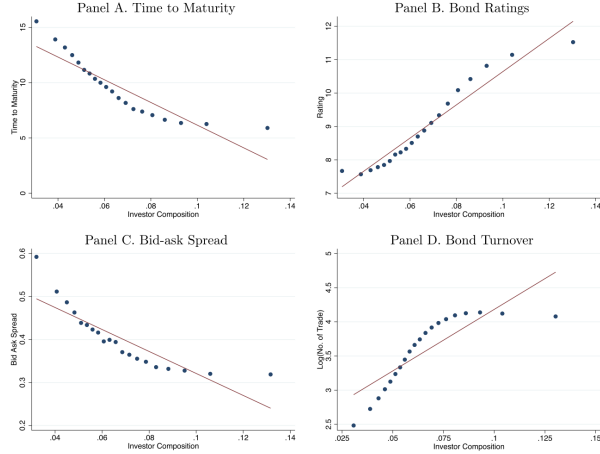


Figure 3. Correlation between bond characteristics and investor types. We sort bonds between 2005Q2 and 2019Q2 according to their investor composition. Note that a higher investor composition measure indicates that the bond is held by shorter term investors. We then plot the average time to maturity for each bin in Panel A. We plot the average numeric rating for each bin in Panel B. Panel C plots the average bid-ask spread for each bin and Panel D plots the average number of trades for each bin. The red lines are linear fitted lines. We exclude bonds with ratings below CCC-, and we winsorize investor composition at the 1% level. Bond information comes from WRDS, investor composition information is constructed from eMaxx, and trading information is from TRACE. (Color figure can be viewed at wileyonlinelibrary.com)

Table I
Bond Turnover, Number of Trades, and Investor Composition

This table presents the results using the discontinuity in investor composition at a maturity of 10 years. Columns (1) and (4) present regression results based on equation (6) for the full sample and subsample of bonds with maturity greater than 10 years at the time of issuance. The remaining columns present results based on equation (7) for the full sample and subsample, where investor composition is instrumented by $1_{ttm>10}$. The control $X_{i,t}$ includes time-to-maturity, age, coupon, log amount outstanding, firm total asset, fraction of long-term debt, leverage ratio, profitability, equity price volatility, slope and level of Treasury yields, as well as rating and industry $\times$ date fixed effects. Standard errors are clustered by industry and date. t -statistics are in parentheses. *, **, and *** indicate statistical significance at the 5%, 1%, and 0.1% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Inv Comp	Turnover	No. of Trades	Inv Comp	Turnover	No. of Trades
$1_{ttm>10}$	-0.00581*** (-5.86)			-0.00692*** (-7.24)		
Inv_Comp		24.34*** (6.41)	75.02*** (8.02)		18.76*** (7.99)	61.62*** (9.69)
Time to Maturity	0.00339*** (6.15)	0.0245* (2.29)	0.0141 (0.48)	0.00326*** (8.79)	0.0190** (2.81)	-0.00404 (-0.19)
Coupon	0.00286*** (4.23)	-0.0675*** (-4.24)	-0.264*** (-6.01)	0.00312*** (4.66)	-0.0561*** (-4.42)	-0.238*** (-6.56)
Log(Amount)	0.00441*** (7.66)	-0.0537** (-3.06)	0.834*** (16.02)	0.00416*** (7.81)	-0.0347** (-2.95)	0.871*** (22.94)
Bandwidth	1.699	1.699	1.699	2.282	2.282	2.282
Sample	Full	Full	Full	Subsample	Subsample	Subsample

- The first-stage coefficient on $1_{ttm>10}$ is around -0.0058 in the full sample and -0.0069 in the subsample of bonds originally issued with maturity above 10 years.
- Instrumented investor composition strongly raises turnover and the number of trades. This is the paper's causal evidence that short-term investors increase secondary-market trading activity.

5.4 Table II: short-term investors amplify the pricing of illiquidity

Read:

- Table II is the key cross-sectional pricing result.
- The interaction coefficient between investor composition and bid-ask spreads is large and positive: about **48.65** in the full sample and **55.86** in the subsample.
- The paper notes that a one-standard-deviation increase in investor composition (about 0.03) raises the loading of credit spreads on bid-ask spreads by about **1.44**.
- This result shows that short-term investors do not just change trading quantities; they change how secondary-market frictions are capitalized into yields.

5.5 Figure 4: numerical equilibrium when the risk-free rate falls

Read:

- Figure 4 visualizes Proposition 3. When the risk-free rate declines, the entire schedule of equilibrium investor types shifts toward shorter-term investors.
- At the same time, seller-buyer ratios fall across all submarkets. This captures the idea that every new short-term investor must first buy a bond, while only investors hit by liquidity shocks need to sell.

Table II
Credit Spreads and Bid-Ask Spreads

This table presents the regression results for equation (8). Inv_Comp is the measure of investor composition, where a higher value indicates that investors have a higher liquidity shock frequency (i.e., shorter term investors). We instrument Inv_Comp with $1_{tm>10}$ and $Inv_Comp \times BA$ with $1_{tm>10} \times BA$. The control $X_{i,t}$ includes time-to-maturity, age, coupon, log amount outstanding, firm total assets, fraction of long-term debt, leverage ratio, profitability, equity price volatility, slope and level of Treasury yields, as well as rating, and industry $\times$ date fixed effects. Standard errors are clustered by industry and time. t -statistics are in parentheses. *, **, and *** indicate statistical significance at the 5%, 1%, and 0.1% level, respectively.

	(1) CS	(2) CS	(3) CS	(4) CS
<i>Bid Ask</i>	0.378*** (4.74)	0.687*** (4.86)	0.232*** (5.41)	0.300* (2.17)
<i>Inv Comp</i>	-0.776 (-1.49)	2.036 (1.26)	-1.166* (-2.47)	-2.479 (-0.89)
<i>Inv Comp $\times$ Bid Ask</i>		48.65*** (6.61)		55.86*** (3.93)
Time to Maturity	0.0154*** (8.84)	0.0168*** (9.94)	0.0150*** (9.95)	0.0147*** (8.24)
Coupon	0.0928*** (7.67)	0.0706*** (5.27)	0.0811*** (7.61)	0.0831*** (7.28)
<i>Log(Amount)</i>	-0.0166 (-0.75)	-0.0375 (-1.35)	-0.0363* (-2.33)	-0.0241 (-1.10)
Sample	Full	Full	Subsample	Subsample

- The figure is helpful because it shows that the rise of short-term investors can simultaneously make trading more frequent and selling pressure lower.

5.6 Table III and Table IV: calibration targets and parameter values

Table III
Target Moments and Model Fits

This table presents the targeted empirical moments and model-produced moment values. The calibration is at the annual level. We focus on bonds between 2 and 15 years of maturity, and thus we set $\delta = 1/15$ and $\bar{\delta} = 0.5$. Furthermore, we assume that the ex ante investor types follow an exponential distribution with scale parameter θ and shape parameter α .

Moment	Empirical	Model Simulated Value
Ave. bond turnover in 2005	0.16	0.16
Ave. bond turnover in 2019	0.21	0.21
Coeff of BA spreads in 2005	0.54	0.54
Coeff of BA spreads in 2019	1.2	1.2
Ave. bond CS in 2005	2.1%	3.111%
Ave. bond CS in 2019	1.48%	2.397%
Ave. bond BA spreads in 2005	0.52%	0.405%
Ave. bond BA spreads in 2019	0.35%	0.343%
BA std./CS std. in 2005	0.44	0.428
BA std./CS std. in 2019	0.31	0.327

Table IV
Model Parameter Values

This table presents the calibrated model parameter values. The calibration is at the annual level. We focus on bonds between 2 and 15 years of maturity, and thus we set $\delta = 1/15$ and $\bar{\delta} = 0.5$. Furthermore, we assume that the ex ante investor types follow an exponential distribution with CDF $1 - \exp(-\alpha(\theta - \underline{\theta}))$.

Parameter	Value
Liquidity discount Δ	0.078
Power parameter in matching function γ	0.806
Scale parameter in matching function η	0.329
Shape parameter of initial distribution α	0.738
Scale parameter of initial distribution $\underline{\theta}$	0.150
Cutoff threshold in 2005 $\bar{\theta}_1$	0.688
Cutoff threshold in 2019 $\bar{\theta}_2$	0.766
Measure of investor flows m_I	0.583
Share of surplus extracted by dealers κ	0.057
Time discount ρ	0.073
Mean of the exo. component in BA spreads μ_e	1.2×10^{-6}
Std. of the exo. component in BA spreads σ_e	9.21×10^{-5}

Read:

- Table III shows that the model is disciplined by concrete moments rather than qualitative patterns only.
- The model exactly targets average turnover rates of **0.16** in 2005 and **0.21** in 2019, and loading coefficients of **0.54** in 2005 and **1.2** in 2019.
- The model overshoots average credit spreads somewhat, but it still matches the relative rise in the liquidity component reasonably well.
- Table IV gives intuitive structural parameters. Dealers extract about **5.7%** of trade surplus ($\kappa = 0.057$), the liquidity-shock discount is about **7.8%**, and the bond-market participation cutoff rises from about **0.688** in 2005 to **0.766** in 2019.

5.7 Table V and the seller-buyer-ratio mechanism

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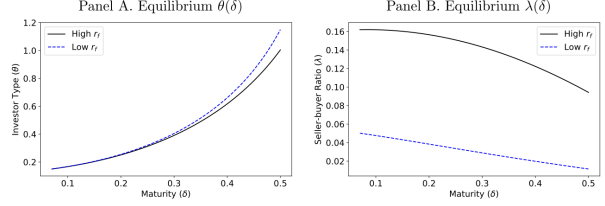


Figure 4. A numerical example of the equilibrium. This figure plots numeric examples of the equilibrium when the risk-free rate is high $r_f = 6.5\%$ (solid black line) and when the risk-free rate is low $r_f = 6\%$ (dotted blue line). Note that a higher value of δ means shorter maturity. We parameterize the distribution of investor type θ with an exponential distribution: $F(\theta) = 1 - \exp(-\alpha(\theta - \underline{\theta}))$. We set $\epsilon_t = 0$ and $d = 0$ for all bonds. The remaining parameter values are: $\Delta = 0.3$, $\gamma = 0.6$, $\eta = 1$, $\underline{\theta} = 0.15$, $m_I = 0.56$, $\kappa = 0.1$, $\alpha = 0.72$, and $\rho = 0.06$. (Color figure can be viewed at wileyonlinelibrary.com)

- Table V is the paper’s main decomposition exercise. It separates the total loading of credit spreads on bid-ask spreads into an exogenous direct piece and an endogenous indirect piece.
- In 2005, the total loading is **0.54**, but the purely exogenous component is only **0.27**. In 2019, the total loading is **1.2**, while the exogenous piece is only **0.35**.
- This means the indirect endogenous-liquidity channel is quantitatively crucial. The correlation between bid-ask spreads and seller-buyer ratios does a lot of the work.
- Panel B shows that both bid-ask spreads and credit spreads load positively on seller-buyer ratios, and these loadings are much larger in the 2019 calibration.

Table V
Decomposing Model Mechanisms

Panel A compares the β coefficient from regression (32) and β_{exo} from regression (34) using model simulated data under the parameterization of the 2005 economy and that of the 2019 economy, respectively. Panel B shows the loading of credit spreads and bid-ask spreads on the seller-buyer ratio in the regression using model simulated data: CS_i or $BA_i = \beta_0 + \beta_\lambda \lambda + \beta_M \delta_i + \varepsilon_i$.

Panel A. Comparing the Effect from the Exogenous Component and the Overall Effect		
	In 2005, $\hat{\theta}_1 = 0.688$	In 2019, $\hat{\theta}_2 = 0.766$
β from equation (32)	0.54	1.2
β_{exo} from equation (34)	0.27	0.35
β_{endo} from equation (35)	0.65	1.7
Empirically observed loading in front of BA	0.54	1.2

Panel B. Effects of Seller-Buyer Ratio		
	In 2005, $\hat{\theta}_1 = 0.688$	In 2019, $\hat{\theta}_2 = 0.766$
Loading of BA on λ	0.0252	0.0557
Loading of CS on λ	0.0212	0.0947

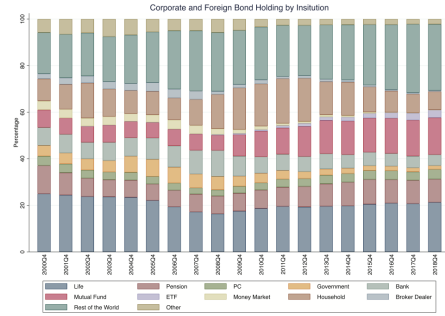


Figure A1. Corporate and foreign bond market breakdowns. *Source:* Flow of Funds. This figure plots the quarterly share of corporate and foreign bond holdings by investor type from the flow of funds. The sample period is 2000Q4 to 2018Q4. “Life” stands for life insurance companies and “PC” stands for property and casualty insurance companies. (Color figure can be viewed at wileyonlinelibrary.com)

5.8 Appendix figures A1, A2, A6, and A7: the background and supporting mechanisms



Figure A2. Number of quarterly trades over time. Panel A plots the median, 25 percentile, and 75 percentile of the number of trades over time. Panel B plots the cross-sectional standard deviation of trades. We exclude retail trades, agency trades, and inter-dealer trades. We exclude the top 1% extreme observations. (Color figure can be viewed at wileyonlinelibrary.com)

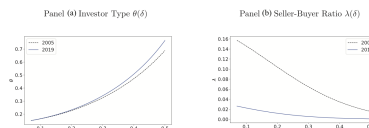


Figure A6. Equilibrium under calibrated parameters. Panel A plots the equilibrium investor type for each submarket under the 2005 parameterization and under the 2019 parameterization in Section IV. Panel B plots the equilibrium seller-buyer ratio. The 2005 economy has a higher risk-free rate and the 2019 economy has a lower risk-free rate. Note that a higher δ corresponds to shorter-maturity bonds. (Color figure can be viewed at wileyonlinelibrary.com)

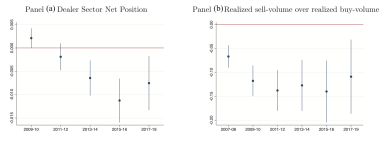


Figure A7. Seller-buyer ratio over time. This figure plots the coefficients on the time fixed effects in equation (A2). Panel A uses the dealer-sector net position to approximate the seller-buyer ratio. To minimize the bias from not observing the initial position of the dealer sector, we exclude observations less than 12 quarters from the first observation of the bond. The sample therefore starts in 2008. In Panel B, we simply use the realized customer-sell volume over the realized customer-buy volume to approximate the seller-buyer ratio. (Color figure can be viewed at wileyonlinelibrary.com)

Read:

- Figure A1 shows the changing ownership structure of the corporate bond market: mutual fund and ETF shares rise substantially over time.
- Figure A2 shows that both the median number of trades and the cross-sectional dispersion of trades increase over time.
- Figure A6 shows that the 2019 equilibrium features shorter-term investors and lower seller-buyer ratios than the 2005 equilibrium.
- Figure A7 shows that empirical proxies for seller-buyer ratios decline over time, consistent with the model’s prediction that the inflow of short-term investors relaxes selling pressure even while making prices more sensitive to frictions.

6 Short summary

- **Method.** The paper combines reduced-form evidence with a search-theoretic model of heterogeneous investors and heterogeneous bonds in an OTC market.
- **Identification.** The aggregate trend comes from repeated cross-sectional regressions of credit spreads on bid-ask spreads; the causal role of investor composition is identified from the discontinuity at the 10-year maturity threshold induced by mutual fund mandates.
- **Mechanism.** As the risk-free rate declines, more short-term investors enter corporate bonds, trade more often, and make credit spreads more sensitive to transaction costs and trade delays.
- **Main result.** The rise of mutual funds and ETFs can quantitatively explain much of the increase in the liquidity component of corporate bond spreads between 2005 and 2019.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that the corporate bond market became more fragile in a particular sense: credit spreads became much more sensitive to secondary-market frictions.
- It also shows that this change can be understood through investor composition. A market populated by shorter-term investors prices transaction costs more heavily, even if measured bid-ask spreads are not rising.
- The calibrated model is able to quantitatively account for the joint evolution of turnover, liquidity sensitivity, and the liquidity component of spreads.

7.2 Economic intuition

- Bid-ask spreads are only one part of the story. What matters for bond pricing is how often those trading frictions are expected to be incurred before maturity.
- When more mutual funds and ETFs hold corporate bonds, bonds trade more often. The marginal investor is more sensitive to both direct transaction costs and indirect waiting costs in the OTC market.
- At the same time, the inflow of investors lowers seller-buyer ratios, so the market can look liquid along some dimensions while still becoming more exposed to liquidity shocks in pricing terms.

7.3 Takeaway

This paper is important because it shifts attention from the *level* of liquidity to the *pricing of liquidity*. In the U.S. corporate bond market, the growth of short-term investors changed that pricing dramatically. As a result, firms' borrowing costs became more exposed to secondary-market frictions, even though standard liquidity indicators alone would not have revealed the full increase in fragility.